

EVE310 Lab 8

October 12, 2023

Announcements

- No lab quiz this week
 - There will be a lab quiz next week that incorporates **both** Labs 8 and 10 in two weeks

Learning objectives

- Logical Operators
- If Statements
- For Loops

If Statement

The Traffic Light: An Everyday IF Statement

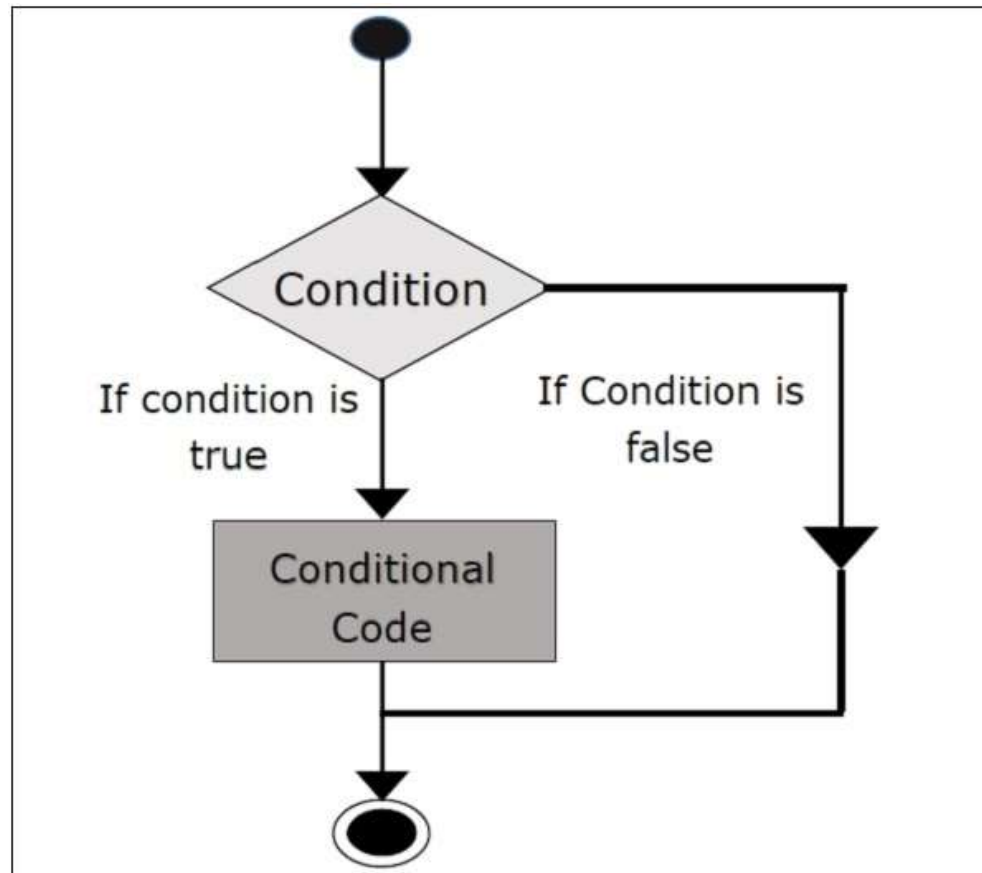


IF light is **red**, THEN stop!

IF light is **yellow**, THEN what???

IF light is **green**, THEN go!

Diagram of an If Statement



Logical Operators

- Statements which ask a true or false question
- Usually evaluate the relationship between two numbers
- True means yes
- False means no

```
a = 5  
b = 4 } Assign values  
  
#Equal to  
print(a == b) } Asking Python: Is a equal to b?
```

False } The logical “truth value” of the statement

Common Logical Operators

- The code snippet to the right contains other common logical operators

```
a = 5
b = 4

#Equal to
print(a == b)

#Greater than
print(a > b)

#Greater than or equal to
print(a >= b)

#less than
print(a < b)

#less than or equal to
print(a <= b)

#Not equal to
print(a != b)
```

Beware the difference between = and ==

- `a = 5` and `b = 4` assigns the value of 5 to `a` and 4 to `b`
- `a == b` asks if the two variables are equal
- In general, `=` assigns a value, and `==` asks if two values are equal
- You do not need to always print a logical statement. Done here to see the output but most times, printing is not necessary.

```
a = 5
b = 4

#Equal to
print(a == b)

#Greater than
print(a > b)

#Greater than or equal to
print(a >= b)

#less than
print(a < b)

#less than or equal to
print(a <= b)

#Not equal to
print(a != b)
```


If Statements

- Execute some code if a logical statement (that you specify) is true

```
#If Statements
a = 3
b = 2
#If A is greater than B, print something to the screen
if (a > b):
    #Code indented here is included in the if statement
    print('The code in the if statement has run')
#This code here is not included in the if statement
print('This is not in the if statement so it will run regardless of the values of a and b')
```

The code in the if statement has run
This is not in the if statement so it will run regardless of the values of a and b

"if" to initiate the if statement

} Assign values

A logical statement inside parenthesis followed by a ":"

Indented code

} Code to be executed if the logical statement is true (note that it is indented)

Code that is not included in the if statement

If-Else Statement

```
a = 3
b = 2

if (b > a):
    #This executes if the conditional statement is true
    print('B is greater than A')

else:
    #This executes if the conditional statement is false
    print('B is less than A')
```

B is less than A

For Loop

- Useful for tasks that are repetitive
- Can condense many lines of code into very simple code
- Can have loops within loops
- We'll start with some very simple tasks, and then work up to more complex ones

Quick Review of Variable Indexing

```
#Create an array  
my_array = np.array([6,5,8,3])
```

What would you type to index the second value of my_array?
(see lab 2)

Quick Review of Variable Indexing

```
#Create an array  
my_array = np.array([6,5,8,3])
```

my_array[1] will return the second value of my_array

Suppose we wanted to print all of the values of my_array

Option 1

```
print('By manually typing in the indices:')
print(my_array[0])
print(my_array[1])
print(my_array[2])
print(my_array[3])
```

Same Result



Option 2

```
print('By changing the value of i, used as the index:')
i=0
print(my_array[i])
i=1
print(my_array[i])
i=2
print(my_array[i])
i=3
print(my_array[i])
```

By manually typing in the indices:

6
5
8
3

By changing the value of i, used as the index:

6
5
8
3

Suppose we wanted to print all of the values of my_array

Option 2

```
print('By changing the value of i, used as the index:')  
i=0  
print(my_array[i])  
i=1  
print(my_array[i])  
i=2  
print(my_array[i])  
i=3  
print(my_array[i])
```

Same Result



Option 3

```
print('By using a for loop:')  
for i in range(0,4):  
    print(my_array[i])
```

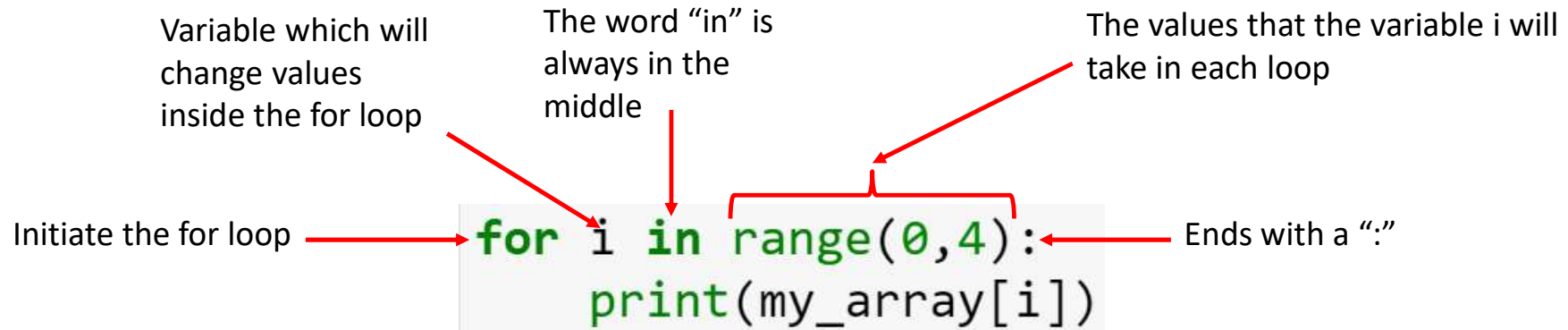
By changing the value of i, used as the index:

6
5
8
3

By using a for loop:

6
5
8
3

For Loop

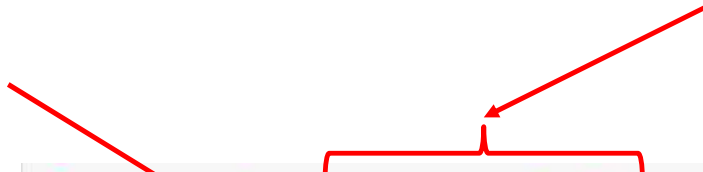


- `range(0,4)` means that I will take the value of each integer from 0 to 3
- Loop will continue until all of the values of `range(0,4)` have been utilized

Another way to think about a for loop

Variable to be
changed inside the
loop

Instructions for how the variable i
should change



```
for i in range(0,4):  
    print(my_array[i])
```

Consider the following examples

```
for i in range(0,4):  
    print(my_array[i])
```

Iteration	Value of i
1	0
2	1
3	2
4	3

```
inds=np.array([4,2,1])  
for i in inds:  
    print(i)
```

Iteration	Value of i
1	4
2	2
3	1

Coding Logic

- *Before* you start coding, write out steps or make a flow chart/diagram *on paper*
- Work through the logic of what you want the code to do
- Identify inputs and outputs
- Break tasks up into small steps
- Focus on the overall process and logic (not specific syntax)

For Loop Logic Demonstration

Use a for loop to find the max value of the following array: [6, 5, 8,3]

Using a for loop and if statement to find the max value of an array

```
#Let's see it in practice

#let's demonstrate how a for loop and if statement works to find the max value of the array d
#We know we could do it with np.max, but let's use a for loop instead

#FILL IN THE ... WITH YOUR CODE

#Make an array
my_array = np.array([6,5,8,3])

#Set the max_val to 0 to start
max_val = 0
#Loop through each value of my_array
for i in range(0,...):
    #If a value of my_array is greater than max_val...
    if ... > ... :
        #Then make that value the new value of max_val
        max_val = ...

#By the end of the loop, we have the maximum value of my_array
print('Max value:',max_val)
```

Iteration	Value of i	Value of my_array[i]	Value of max_val
1	0	6	6
2	1	5	6
3	2	8	8
4	3	3	8

Fill in the ... to complete the code!
Be careful of spaces, :, (), [], and indentations

Using a for loop and if statement to find the max value of an array

Solution:

```
#Let's see it in practice

#let's demonstrate how a for loop and if statement works to find the max value of the array d
#We know we could do it with np.max, but let's use a for loop instead

#FILL IN THE ... WITH YOUR CODE

#Make an array
my_array = np.array([6,5,8,3])

#Set the max_val to 0 to start
max_val = 0
#Loop through each value of my_array
for i in range(0,4):
    #If a value of my_array is greater than max_val...
    if my_array[i] > max_val:
        #Then make that value the new value of max_val
        max_val = my_array[i]

#By the end of the loop, we have the maximum value of my_array
print('Max value:',max_val)
```

Max value: 8

Iteration	Value of i	Value of my_array[i]	Value of max_val
1	0	6	6
2	1	5	6
3	2	8	8
4	3	3	8

Online Game to help with intuition of if statements/for loops

<https://compute-it.toxicode.fr/?hour-of-code&progression=python>

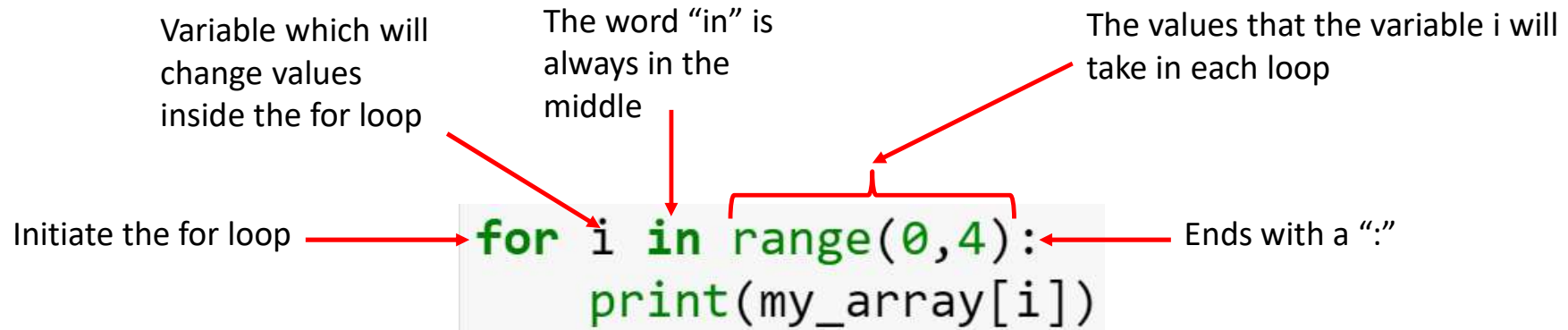


In-class activity

- Work through Lab 8 Student Activity
- Tip for activity: Don't immediately start typing code. Take a moment to think about what you are trying to do and how to do that. Then type some code.
- **Mid-Semester Survey for TA Certification (Thank you!):**
https://utexas.qualtrics.com/jfe/form/SV_a9jfRgUQG8PGAGW?TA=kk35355

If Statement Demonstration

For Loop



The `range(0,4)` means that I will take the value of each integer from 0 to 3